Zihao (Gavin) Yang

Boston, MA | 646-897-9271 | yang.zih@northeastern.edu | redagavin.github.io | Google Scholar | LinkedIn

RESEARCH INTERESTS

I study whether large language models' stated reasoning (chain of thought) faithfully reports what drives their answers, and how to measure that faithfulness and train for it. I build counterfactual and interpretability-based evaluations with matched baselines, and train model organisms with hidden influences as ground truth. My research is motivated by the need for trustworthy, auditable AI, especially in high-stakes domains. I am available for internships from May 2027 to August 2027.

EDUCATION

Northeastern University — Boston, MA

Ph.D. in Computer Science; Advisor: Prof. Byron C. Wallace — *Sept 2025 – May 2030 (expected)*

New York University — New York, NY

B.A. in Computer Science and Data Science, Minor in Mathematics; GPA 3.82/4.0 — *Sept 2020 – May 2024*

AWARDS & HONORS

- George Maker Research Scholar, New York University, 2023
- Dean's Undergraduate Research Fund Grant, New York University, 2023
- Dean's List for Academic Year, New York University, 2021–2022, 2022–2023, 2023–2024
- Cum Laude, New York University, 2024

PEER-REVIEWED PUBLICATIONS

[1] **Z. Yang**, M. Levy, Y. Goldberg, B. C. Wallace. Compared to What? Baselines and Metrics for Counterfactual Prompting. ***COLM***, 2026.

[2] Y. Chen, R. Wang, **Z. Yang**, L. Y. Jiang, E. K. Oermann. Optimizing Packing and Shuffling Strategies for Enhanced Performance in Generative Language Models. ***ACL*** *SRW*, 2026.

[3] S. Yang*, J. Yang*, P. Huang†, E. Brown†, **Z. Yang**, Y. Yu, S. Tong, Z. Zheng, Y. Xu, M. Wang, R. Fergus, Y. LeCun, L. Fei-Fei, S. Xie. Cambrian-S: Towards Spatial Supersensing in Video. ***ICLR***, 2026. (* equal contribution; † core contributor)

[4] D. Alber, **Z. Yang**, A. Alyakin, E. Yang, S. Rai, A. Valliani, J. Zhang, G. Rosenbaum, A. Amend-Thomas, D. Kurland, C. Kremer, A. Eremiev, B. Negash, D. Wiggan, M. Nakatsuka, K. Sangwon, S. Neifert, H. Khan, A. Save, A. Palla, E. Grin, M. Hedman, M. Nasir-Moin, X. Liu, L. Jiang, M. Mankowski, D. Segev, Y. Aphinyanaphongs, H. Riina, J. Golfinos, D. Orringer, D. Kondziolka, E. Oermann. Medical large language models are vulnerable to data-poisoning attacks. ***Nature Medicine***, 2025.

[5] **Z. Yang**, C. Zhang, M. Wu, X. Liu, L. Jiang, K. Cho, E. Oermann. Intriguing Effect of the Correlation Prior on ICD-9 Code Assignment. ***ACL*** *SRW*, 2023.

PREPRINTS & MANUSCRIPTS

[6] C. Hang, R. Deng, L. Jiang, **Z. Yang**, A. Alyakin, D. Alber, E. Oermann. BPQA Dataset: Evaluating How Well Language Models Leverage Blood Pressures to Answer Biomedical Questions. arXiv:2503.04155, 2025.

[7] L. Jiang, D. Alber, **Z. Yang**, K. Sangwon, X. Liu, K. Cho, E. Oermann. Language Models Can Guess Your Identities from De-identified Clinical Notes. Under review.

RESEARCH EXPERIENCE

Faithfulness Beyond Counterfactual Simulatability [Ongoing] — Boston, MA

Northeastern University, Advisor: Prof. Byron C. Wallace — *Aug 2026 – Present*

- Built a model organism: Qwen3-4B fine-tuned to follow answer hints from 28 planted sources while verbalizing in its CoTs only the hint types that Counterfactual Simulation Training (CST) inserts.
- Showed that, in CST's setup, the organism scores 0.76 ± 0.01 on CST's simulatability score, the level CST's

authors reach after their own training, yet follows 94–99% of hidden hints and mentions them in about 2% of its CoTs; CST's own training raised the score (0.78 to 0.83) without reducing the hidden behavior.

Baselines and Metrics for Counterfactual Prompting [COLM 2026] Boston, MA

Northeastern University, Advisor: Prof. Byron C. Wallace *Oct 2025 – Aug 2026*

- Showed that counterfactual prompt edits are compound treatments: on MedQA, changing patient gender flips answers 14.9% of the time, statistically indistinguishable from the 14.1% from paraphrasing alone.
- Proposed an edit-length-matched paraphrase baseline with statistical testing: only 5 of 120 tests of MedPerturb's reported demographic and stylistic effects stay significant against it, while directional gender bias in Bias-in-Bios is detected even when small; per-sample metrics (JSD, KL) prove far more powerful than aggregate ones (MI, φ, flip rate). Released the cfprompt Python package.

Which Influences Do Reasoning Models Admit? [Ongoing] Boston, MA

Northeastern University, Advisor: Prof. Byron C. Wallace *May 2026 – Present*

- Designed eight answer hints that vary one property at a time (source legitimacy, or the act performed, such as bribe or threat) and ran them on DeepSeek-R1-Distill-Llama-70B: a hint's source (peer-reviewed study vs leaked answer key) does not change how often the CoT admits it (0.82–0.88), but the kind of hint does, fourfold (professor 0.88, bribe 0.22); inappropriate hints are followed and admitted less.
- Found that hint wording alone swings measured faithfulness from 14% to 100%, and that in about 1 in 10 generations the answer is not decided in the CoT and cannot be explained from it alone.

Language-Supervised Vision Models and Efficient Multimodal LLMs New York, NY

NYU VisionX Lab, Advisor: Prof. Saining Xie *July 2024 – June 2025*

- **Self-improving vision representations:** an MLLM generates billion-scale VQA data to train its own vision encoder, for better vision representations and task generalizability than CLIP; built an open-vocabulary detection pipeline (OwlViTv2, RAM++) for VQA generation and pretrained ViT baselines with PaliGemma and Cambrian on DataComp-Recap.
- **Efficient MLLMs:** investigated vision-token compression (Token Merging) and model merging (RegMean) to compress MLLMs for efficient on-device inference; migrated the Cambrian codebase from TPU to GPU.
- **Cambrian-S (ICLR 2026):** co-authored; led model evaluation across its benchmarks and datasets.

Safety, Training, and Evaluation of Medical Language Models New York, NY

NYU Langone OLAB, Advisors: Prof. Eric K. Oermann; Prof. Kyunghyun Cho *June 2022 – July 2024*

- **Data poisoning (Nature Medicine, 2025):** replacing only 0.001% of pretraining tokens with medical misinformation significantly increases harmful outputs while leaving performance on medical benchmarks unchanged, so the attack is undetectable; built a validation method using named-entity recognition and vector-similarity search on biomedical knowledge graphs that catches 91.9% of harmful content.
- **Pretraining dynamics:** toward a trustworthy medical LLM, built a LitGPT/PyTorch Lightning framework and trained multiple 1B models on Starcoder, PubMed abstracts, and SlimPajama mixtures to measure how data mix affects downstream performance; in-domain PubMed data improves clinical fine-tuning.
- **Neurosurgical benchmark:** built a multiple-choice benchmark from two neurosurgical board-exam datasets; most state-of-the-art LLMs score up to 42% lower on it than on five general medical QA datasets.
- **ICD-9 code assignment (ACL 2023 SRW, first author):** predicting diagnosis and procedure codes from clinical text; proposed a prior from correlations between ICD-9 and other medical codes, passively (training on two other code types) and actively (a loss regularizer), which improves performance on specific codes.

SERVICE

Reviewer: ICLR 2025, 2026; COLM 2026; NeurIPS 2026 (Evaluations and Datasets Track)

SKILLS

Python, PyTorch, JAX, TensorFlow, Hugging Face, Java, C, C++; LLM post-training (SFT, DPO, RL, LoRA); multi-node GPU training & inference; interpretability (TransformerLens, NNsight, SAEs, probing, J-lens)